

SMART INDIA HACKATHON 2026



- Problem Statement ID –26167
- Problem Statement Title- SatQuery AI - AN INTERACTIVE VISION-LANGUAGE ASSISTANT FOR MULTIMODAL REMOTE SENSING IMAGE ANALYSIS THROUGH TEXT QUERIES
- Theme- SPACE TECHNOLOGY
- PS Category- SOFTWARE
- Team ID- 179687
- Team Name(Registered on portal) – VisionX



SatQuery

PROBLEM ?

Hard to Understand -

Satellite images contain complex information that is difficult to interpret for nonexpert.

Different Information -

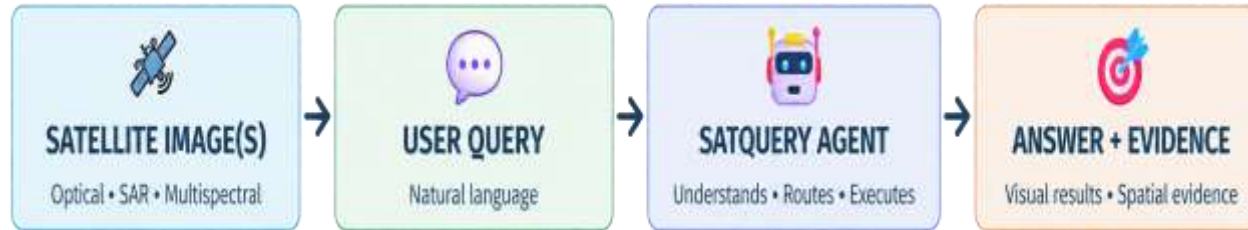
Optical and SAR images reveal different details about the same area.

Change Needs Multiple Images -

Changes over time require comparing images from different dates.

Too Many Tools -

Different questions often require **different** models and analysis tools.



Solution

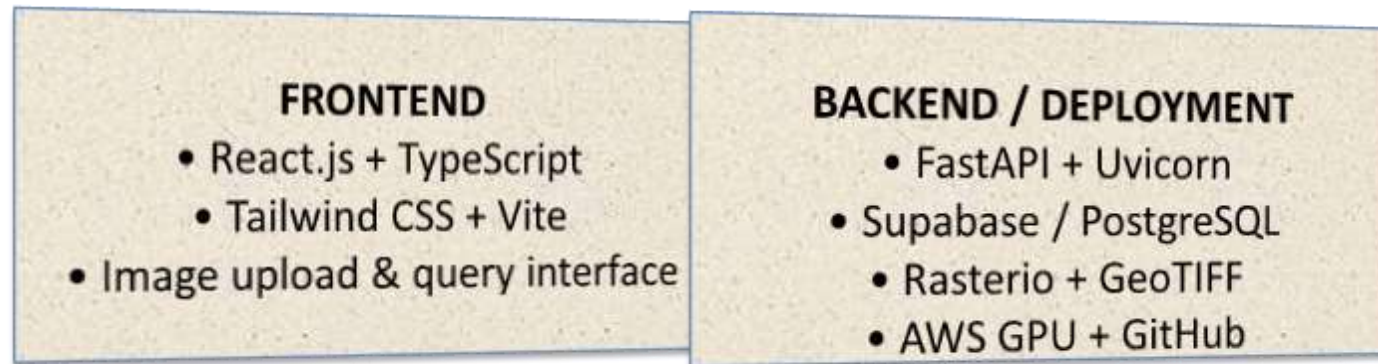
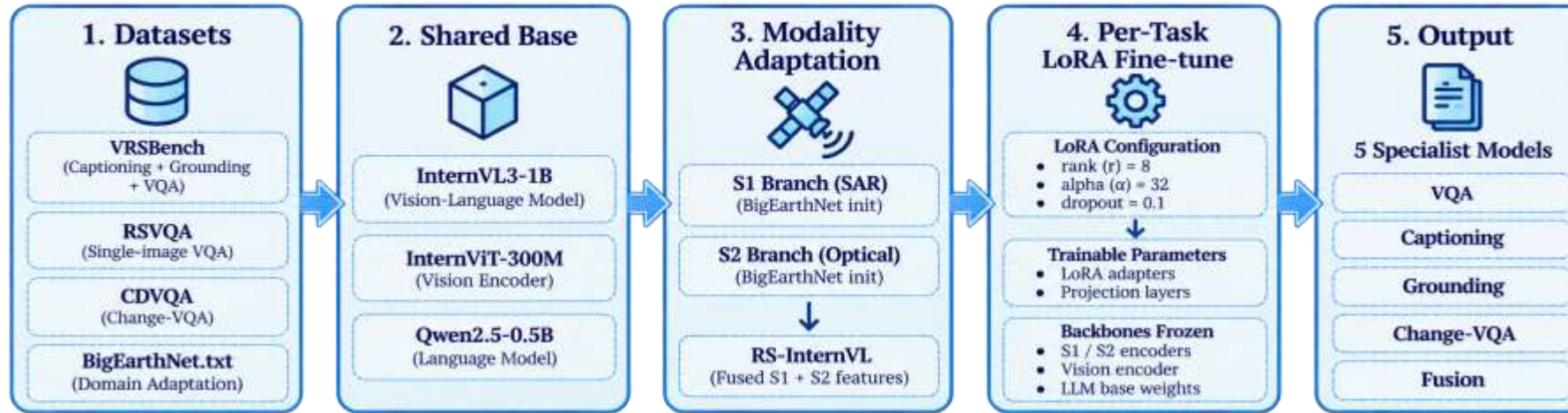
- Ask questions in natural language
- Analyzes optical, SAR, single-image and multi-temporal imagery in one workflow.
- Automatically select the right AI task
- Return answers with visual evidence

WHY IS SATQUERY DIFFERENT?

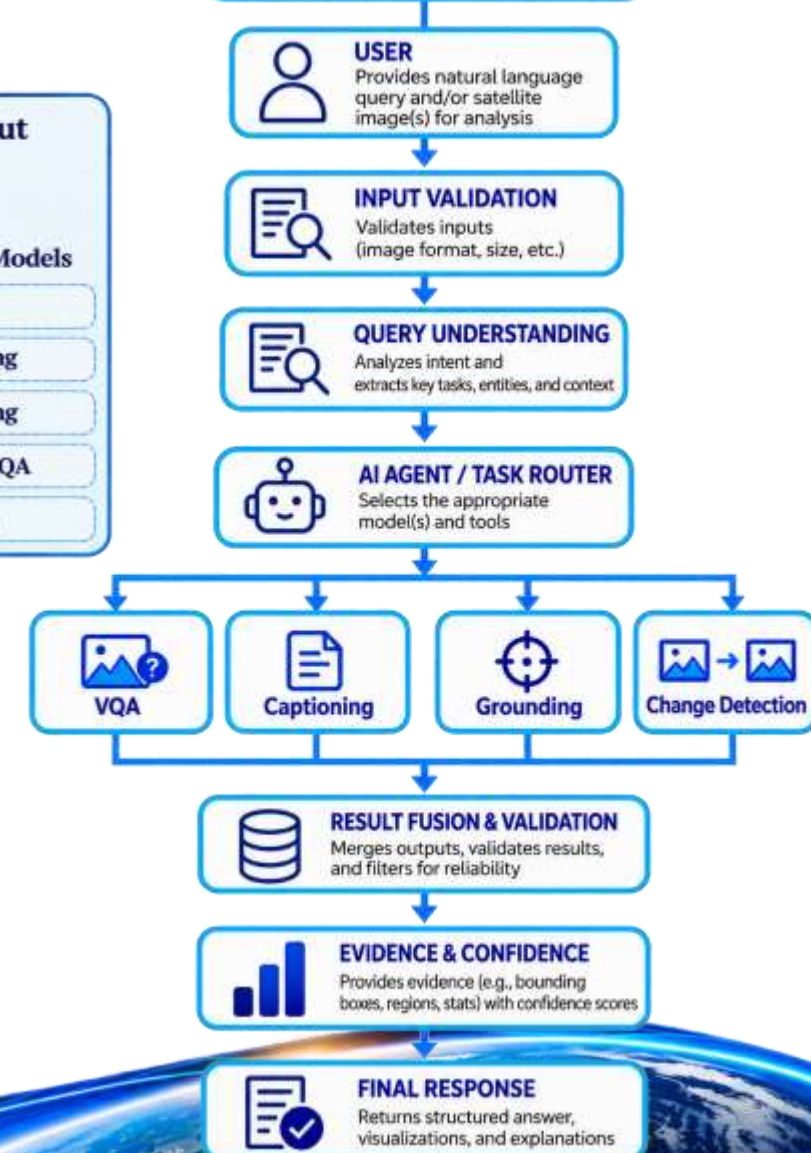
- **NATURAL LANGUAGE**
Ask questions about satellite images in simple language.
- **AGENTIC AI**
Understands the query → plans → selects the right models → executes the workflow.
- **MULTIMODAL**
Combines Optical + SAR information for richer analysis.
- **MULTI-TEMPORAL**
Compares images from different times to understand change.
- **EVIDENCE-GROUNDED**
Provides answers with confidence scores and visual/spatial evidence.
- **COMPLEX → SIMPLE**
Converts complex satellite analysis into easy-to-understand, actionable answers.

 [YouTube Demo Link](#)

Training / Adaptation Pipeline



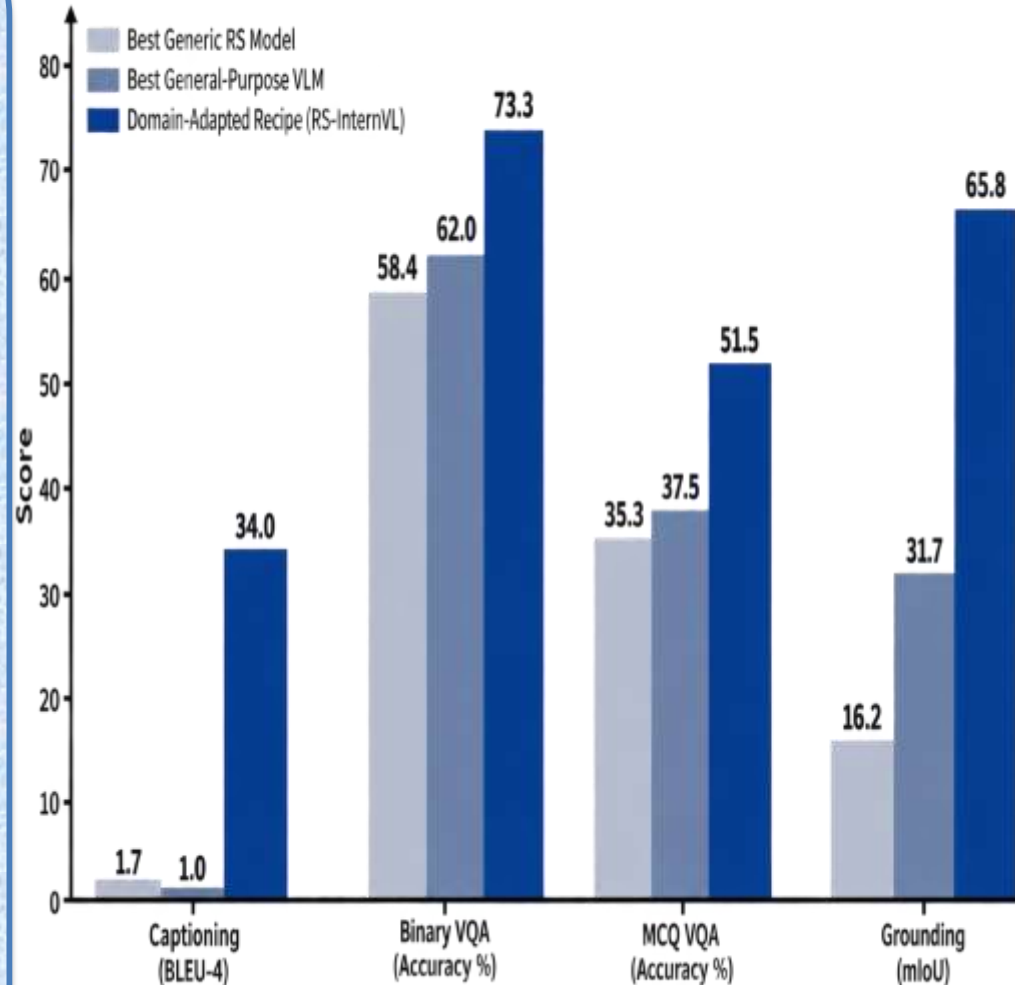
SYSTEM ARCHITECTURE



FEASIBILITY ANALYSIS

- **Foundation Model Reuse:**
Uses InternVL3-1B with lightweight LoRA adaptation instead of training from scratch.
- **Validated Methodology:**
Published RS-InternVL results show strong gains on BigEarthNet.txt benchmarks.
- **Modular Development:**
Task-specific adapters support VQA, Captioning, Grounding, Change Analysis and Fusion.
- **Compute Efficient:**
LoRA reduces trainable parameters, making domain adaptation practical on research-grade GPUs.

Why This Approach: Domain-Adapted Methodology vs. Generic Models
(Published Benchmark, BigEarthNet.txt Evaluation Set)



Reference: BioEarthNet.txt paper (arXiv 2603.29630, TU Berlin/BIFOLD) – the published methodology our RS-adaptation pipeline is built on. Own model results to be reported separately upon completion.

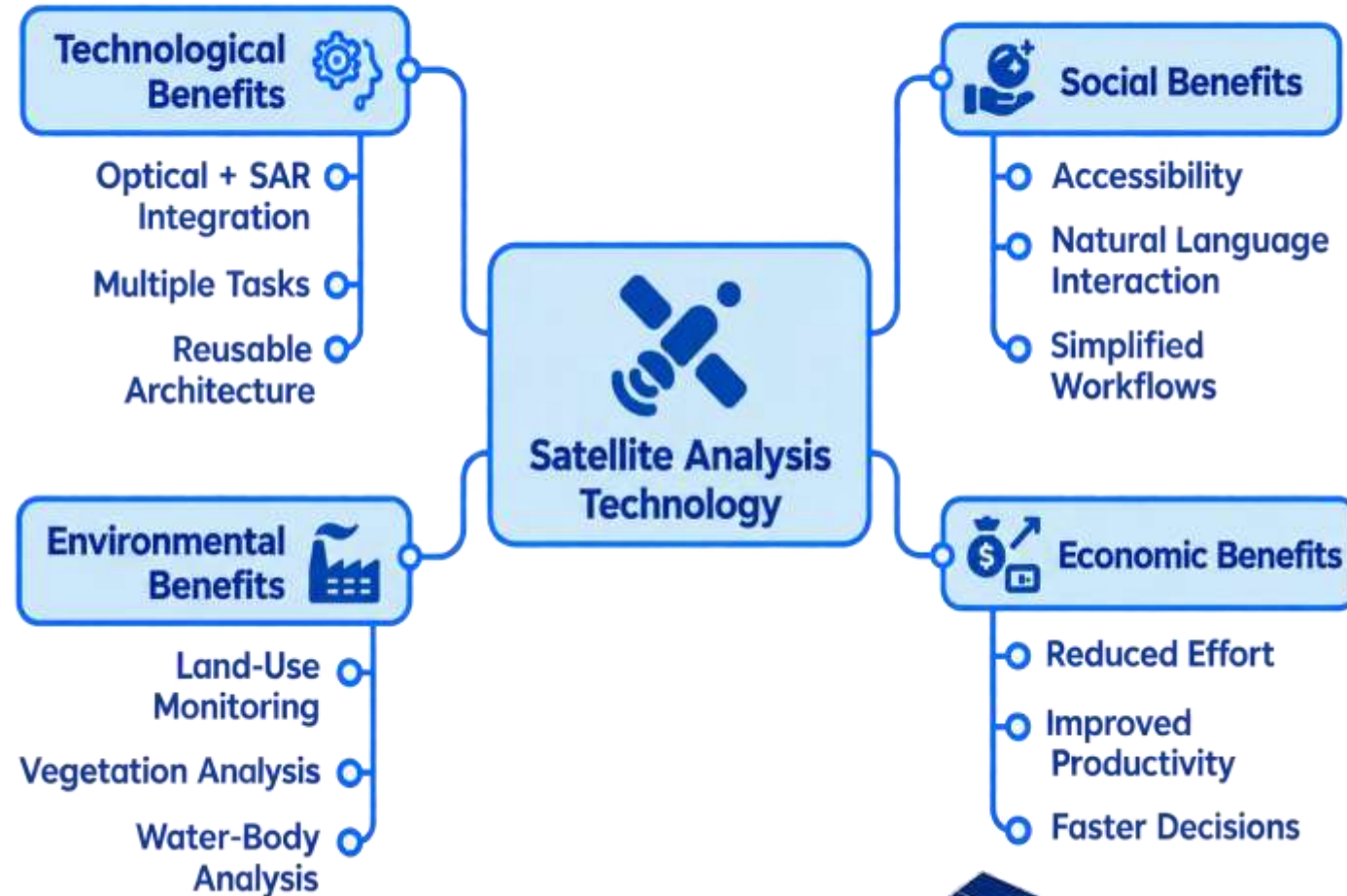
VIABILITY

- **Open-Source Ecosystem:**
Uses open models, datasets and tools, reducing dependency on costly external APIs.
- **Multi-Modal Analysis:**
Supports Optical, SAR and multi-temporal satellite imagery.
- **Evidence-Based Results:**
Provides confidence scores with visual and spatial evidence.
- **Real-World Applications:**
Applicable to disaster response, agriculture, land-use and infrastructure.

IMPACT AND BENEFITS

TARGET AUDIENCE & IMPACT

- **GIS & REMOTE-SENSING PROFESSIONALS**
Impact: Faster exploration and analysis of complex satellite imagery.
- **GOVERNMENT & PLANNING AGENCIES**
Impact: Supports faster monitoring and evidence-based planning.
- **AGRICULTURE & LAND MANAGEMENT**
Impact: Enables easier analysis of land-use, vegetation and spatial changes.
- **RESEARCHERS & STUDENTS**
Impact: Simplifies experimentation and development with satellite data.



VRSBench
NeurIPS 2024
Research Paper Link

BigEarthNet.Txt
IEEE IGRASS 2019
Research Paper Link

CDVQA
IEEE TGRS 2022
Research Paper Link

 Docker Deployment

Pull:

`docker pull ghcr.io/afsar426/satquery-ai:sih-final`

Run:

`docker run --rm --gpus all -p 8080:8080 ghcr.io/afsar426/satquery-ai:sih-final`

Open:

`http://localhost:8080`

RSVQA
IEEE 2020
Research Paper Link

SkySense
IEEE 2025
Research Paper Link

SIH26167
Smart India Hackathon 2026




Prototype

